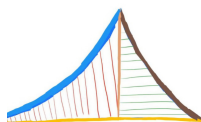


Math for EE

1



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ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

CONTENTS

1	GATE AE 2026	4
1.1	Formulae	4
1.2	Problems	16

1. GATE AE 2026

A. Formulae

1.1.1

$$\Pr(M) + \Pr(E) - \Pr(ME) + \Pr(M + E)' = 1 \quad (1.1.1.1)$$

1.1.2

$$A + A' = 1, \quad AA' = 0 \quad (1.1.2.1)$$

$$\implies \Pr(A + A') = \Pr(1) = 1 \quad \Pr(AA') = \Pr(0) = 0 \quad (1.1.2.2)$$

1.1.3

$$\Pr(A + A') = \Pr(A) + \Pr(A') \quad (1.1.3.1)$$

1.1.4 Difference rule of probability

$$\Pr(AB') = \Pr(A) - \Pr(AB) \quad (1.1.4.1)$$

Solution: From (1.1.2.1)

$$A = A \cdot 1 = A(B + B') = AB + AB' \quad (1.1.4.2)$$

Also,

$$(AB)(AB') = 0 \quad (1.1.4.3)$$

From (1.1.3.1),

$$\Pr(A) = \Pr(AB + AB') = \Pr(AB) + \Pr(AB') \quad (1.1.4.4)$$

Rearranging, we get:

$$P(AB') = P(A) - P(AB) \quad (1.1.4.5)$$

1.1.5 The Laplace transform is defined as

$$g(t) \xleftrightarrow{\mathcal{L}} G(s) = \int_0^\infty e^{-st} g(t) dt \quad (1.1.5.1)$$

1.1.6

$$g'(t) \xleftrightarrow{\mathcal{L}} sG(s) - g(0) \quad (1.1.6.1)$$

Solution: Applying integration by parts,

$$u = e^{-st}, dv = g'(t) dt \quad (1.1.6.2)$$

$$\implies \int_0^\infty e^{-st} g'(t) dt = [e^{-st} g(t)]_0^\infty - \int_0^\infty (-se^{-st}) g(t) dt \quad (1.1.6.3)$$

$$= -g(0) + s \int_0^\infty e^{-st} g(t) dt \quad (1.1.6.4)$$

yielding (1.1.6.1).

1.1.7

$$g''(t) \xleftrightarrow{\mathcal{L}} s^2 G(s) - sg(0) - g'(0) \quad (1.1.7.1)$$

1.1.8

$$e^{at}g(t) \xleftrightarrow{\mathcal{L}} G(s-a) \quad (1.1.8.1)$$

Solution:

$$\Rightarrow e^{at}g(t) \xleftrightarrow{\mathcal{L}} \int_0^\infty e^{-st} (e^{at}g(t)) dt \quad (1.1.8.2)$$

$$= \int_0^\infty e^{-(s-a)t} g(t) dt \quad (1.1.8.3)$$

yielding (1.1.8.1).

1.1.9 The unit step function is defined as

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases} \quad (1.1.9.1)$$

1.1.10

$$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s} \quad (1.1.10.1)$$

Solution:

$$\int_0^\infty e^{-st} dt = \left(-\frac{e^{-st}}{s} \right)_0^\infty = \frac{1}{s} \quad \Re(s) > 0 \quad (1.1.10.2)$$

1.1.11

$$tu(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s^2} \quad (1.1.11.1)$$

Solution: Differentiating (1.1.10.2) with respect to s results in (1.1.11.1).

$$\begin{aligned} \cos(at)u(t) &\xleftrightarrow{\mathcal{L}} \frac{s}{s^2 + a^2} \\ \sin(at)u(t) &\xleftrightarrow{\mathcal{L}} \frac{a}{s^2 + a^2} \end{aligned} \quad \Re(s) > a \quad (1.1.11.2)$$

Solution: From (1.1.8.1) and (1.1.10.1),

$$e^{jat}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s - ja}, \quad \Re(s) > a \quad (1.1.11.3)$$

$$\Rightarrow \cos(at)u(t) + j \sin(at)u(t) \xleftrightarrow{\mathcal{L}} \frac{s + ja}{(s - ja)(s + ja)} \quad (1.1.11.4)$$

$$= \frac{s}{s^2 + a^2} + j \frac{a}{s^2 + a^2} \quad (1.1.11.5)$$

resulting in (1.1.11.2).

1.1.12 Solve

$$\frac{d^2y}{dx^2} = -\alpha^2 y, \alpha \neq 0 \quad (1.1.12.1)$$

Solution: Applying the Laplace transform on both sides, and using (1.1.7.1),

$$s^2 Y(s) - sy(0) - y'(0) = -\alpha^2 Y(s) \quad (1.1.12.2)$$

Substituting

$$y(0) = \beta, y'(0) = \gamma \quad (1.1.12.3)$$

in (1.1.12.2),

$$Y(s) = \frac{s\beta + \gamma}{s^2 + \alpha^2} \quad (1.1.12.4)$$

$$= \frac{s\beta}{s^2 + \alpha^2} + \frac{\gamma}{s^2 + \alpha^2} \quad (1.1.12.5)$$

yielding

$$y(t) = \beta \cos(\alpha t) + \frac{\gamma}{\alpha} \sin(\alpha t) \quad (1.1.12.6)$$

from (1.1.11.2).

1.1.13 Euler's Method of Forward Differences. Numerically solve

$$\frac{dy}{dx} = f(x, y), y(x_0) = y_0 \quad (1.1.13.1)$$

Solution: By definition,

$$\left. \frac{dy}{dx} \right|_{x=x_0} = \lim_{h \rightarrow 0} \frac{y(x_0 + h) - y(x_0)}{h} \quad (1.1.13.2)$$

which can be expressed using the notation

$$y'_0 = \frac{y_1 - y_0}{h} \quad (1.1.13.3)$$

Similarly,

$$y''_n = \frac{y'_{n+1} - y'_n}{h} \quad (1.1.13.4)$$

From (1.1.13.3),

$$y'_{n+1} = \frac{y_{n+2} - y_{n+1}}{h}, y'_n = \frac{y_{n+1} - y_n}{h} \quad (1.1.13.5)$$

yielding

$$y''_n = \frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} \quad (1.1.13.6)$$

upon simplification.

1.1.14 Kronecker delta function

$$\delta_n = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases} \quad (1.1.14.1)$$

1.1.15

$$f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{j2\pi n f_0 t} dt = \delta_n \quad (1.1.15.1)$$

1.1.16 The Fourier Series. Let

$$x(t) = \sum_{k=-\infty}^{\infty} x_0(t)(t - nT_0) \quad (1.1.16.1)$$

be a periodic signal with

$$x(t) = \begin{cases} x_0(t), & t \in \left(-\frac{T_0}{2}, \frac{T_0}{2}\right) \\ 0 & \text{otherwise} \end{cases}. \quad (1.1.16.2)$$

Then,

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_0 t}, \quad f_0 = \frac{1}{T_0} \quad (1.1.16.3)$$

Show that

$$c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0 t} dt \quad (1.1.16.4)$$

Solution: From (1.1.16.3),

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-j2\pi n f_0 t} dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} \left(\sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_0 t} \right) e^{-j2\pi n f_0 t} dt \quad (1.1.16.5)$$

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0 t} dt = \sum_{k=-\infty}^{\infty} c_k \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{j2\pi(k-n)f_0 t} dt \quad (1.1.16.6)$$

$$= \frac{1}{f_0} \sum_{k=-\infty}^{\infty} c_k \delta_{k-n} \quad (1.1.16.7)$$

upon substituting from (1.1.16.2) and (1.1.15.1), yielding (1.1.16.4).

1.1.17 The rectangular function. The unit rectangular function is defined as

$$\text{rect}(t) = \begin{cases} 1 & t \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ 0 & \text{otherwise} \end{cases} \quad (1.1.17.1)$$

1.1.18 Square Wave. Consider the Fourier series with

$$x_0(t) = \text{rect}(t), \quad T_0 = 2. \quad (1.1.18.1)$$

Show that

$$c_n = \frac{\sin\left(\frac{\pi k}{2}\right)}{\pi k} \quad (1.1.18.2)$$

Solution: Substituting $f_0 = \frac{1}{2}$ in (1.1.16.4),

$$c_n = \frac{1}{2} \int_{-1}^1 e^{-j\pi k t} dt \quad (1.1.18.3)$$

$$= \left[\frac{\sin(\pi k t)}{2\pi k} \right]_{-\frac{1}{2}}^{\frac{1}{2}} \quad (1.1.18.4)$$

The following code verifies the result in Fig. 1.1.18.1. Note that Fig. 1.1.18.1 is continuous while (1.1.18.1) is discontinuous. So (1.1.18.2) provides only an approximation. This is known as the Gibbs' phenomenon.

codes/Fourier.py

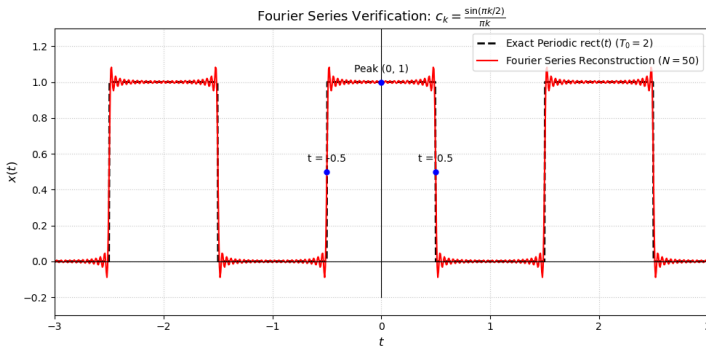


Fig. 1.1.18.1: Gibbs phenomenon

1.1.19 If

$$x(t) \xleftrightarrow{\mathcal{F}} c_n, \quad (1.1.19.1)$$

show that

$$x(t - t_0) \xleftrightarrow{\mathcal{F}} c_n e^{-j2\pi n t_0}, \quad (1.1.19.2)$$

Solution:

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t - t_0) e^{-j2\pi n f_0 t} dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0(t+t_0)} dt \quad (1.1.19.3)$$

$$= e^{j2\pi n f_0 t_0} \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t) e^{-j2\pi n f_0 t} dt \quad (1.1.19.4)$$

after a change of variables. Note that the limits of integration are unchanged due to the periodicity of $x(t)$. Transferring the exponential term in (1.1.19.4) to the L.H.S yields (1.1.19.2).

1.1.20 The standard 2-D basis vectors are defined as

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (1.1.20.1)$$

Also,

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.1.20.2)$$

1.1.21 For

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top, \text{rank}(\mathbf{A}) = 1. \quad (1.1.21.1)$$

where both u and v are n -dimensional real-valued non-zero column vectors.

Solution: Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}. \quad (1.1.21.2)$$

Then

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 4 & 2 \end{pmatrix} \quad (1.1.21.3)$$

$$= \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \quad (1.1.21.4)$$

Following row reduction,

$$\mathbf{A} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 3 & 12 & 6 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (1.1.21.5)$$

yielding (1.1.21.1).

1.1.22

$$|\mathbf{A}^\top| = |\mathbf{A}| \quad (1.1.22.1)$$

1.1.23

$$|\mathbf{AB}| = |\mathbf{A}| |\mathbf{B}| \quad (1.1.23.1)$$

Solution: If $\mathbf{A}$ is invertible

$$\mathbf{A} = \mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k \quad (1.1.23.2)$$

where $\mathbf{E}_i$ are invertible matrices. For any single elementary matrix $\mathbf{E}$, it is easy to

verify that

$$|\mathbf{EB}| = |\mathbf{E}| |\mathbf{B}|. \quad (1.1.23.3)$$

$$\therefore |\mathbf{AB}| = |\mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k \mathbf{B}| \quad (1.1.23.4)$$

$$= |\mathbf{E}_1| |\mathbf{E}_2 \cdots \mathbf{E}_k \mathbf{B}| \quad (1.1.23.5)$$

$$= |\mathbf{E}_1| |\mathbf{E}_2| \cdots |\mathbf{E}_k| |\mathbf{B}| \quad (1.1.23.6)$$

For $\mathbf{B} = \mathbf{I}$,

$$|\mathbf{A}| = |\mathbf{E}_1| |\mathbf{E}_2| \cdots |\mathbf{E}_k| \quad (1.1.23.7)$$

which upon substituting in (1.1.23.6) yields

$$|\mathbf{AB}| = |\mathbf{A}| |\mathbf{B}| \quad (1.1.23.8)$$

1.1.24

$$|\mathbf{A}^{-1}| = \frac{1}{|\mathbf{A}|} \quad (1.1.24.1)$$

1.1.25 If

$$\mathbf{Ax} = \lambda \mathbf{x}, \quad (1.1.25.1)$$

then λ is an eigenvalue $\mathbf{x}$ of $\mathbf{A}$.

1.1.26 Let $\lambda_1, \lambda_2, \dots, \lambda_i$ be the eigenvalues of a matrix $\mathbf{A}$. Then,

$$\sum_{i=1}^n \lambda_i = \text{Tr}(\mathbf{A}) \quad (1.1.26.1)$$

$$\prod_{i=1}^n \lambda_i = |\mathbf{A}| \quad (1.1.26.2)$$

1.1.27 $\mathbf{A}$ and $\mathbf{A}^\top$ have the same eigenvalues.

Solution: From (1.1.24.1),

$$|\mathbf{A} - \lambda \mathbf{I}| = |(\mathbf{A} - \lambda \mathbf{I})^\top| \quad (1.1.27.1)$$

$$= |\mathbf{A}^\top - \lambda \mathbf{I}|, \quad (1.1.27.2)$$

completing the proof.

1.1.28 The equation of a conic with directrix $\mathbf{n}^\top \mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$g(\mathbf{x}) = \mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (1.1.28.1)$$

where

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top, \quad (1.1.28.2)$$

$$\mathbf{u} = ce^2 \mathbf{n} - \|\mathbf{n}\|^2 \mathbf{F}, \quad (1.1.28.3)$$

$$f = \|\mathbf{n}\|^2 \|\mathbf{F}\|^2 - c^2 e^2 \quad (1.1.28.4)$$

Solution:

$$\|\mathbf{x} - \mathbf{F}\|^2 = e^2 \frac{(\mathbf{n}^\top \mathbf{x} - c)^2}{\|\mathbf{n}\|^2} \quad (1.1.28.5)$$

$$\implies \|\mathbf{n}\|^2 (\mathbf{x} - \mathbf{F})^\top (\mathbf{x} - \mathbf{F}) = e^2 (\mathbf{n}^\top \mathbf{x} - c)^2 \quad (1.1.28.6)$$

$$\implies \|\mathbf{n}\|^2 (\mathbf{x}^\top \mathbf{x} - 2\mathbf{F}^\top \mathbf{x} + \|\mathbf{F}\|^2) = e^2 (c^2 + (\mathbf{n}^\top \mathbf{x})^2 - 2c\mathbf{n}^\top \mathbf{x}) \quad (1.1.28.7)$$

$$= e^2 (c^2 + (\mathbf{x}^\top \mathbf{n} \mathbf{n}^\top \mathbf{x}) - 2c\mathbf{n}^\top \mathbf{x}) \quad (1.1.28.8)$$

which can be expressed as (1.1.28.1) after simplification.

1.1.29 The polar form of a conic section is given as:

$$r = \frac{ep}{1 + e \cos(\theta - \theta_0)} \quad (1.1.29.1)$$

where

- r is the distance from the focus(origin) to a point on the curve.
- e is the eccentricity of the conic
- p is the distance from the focus to the directrix
- θ_0 is the angle that defines the orientation of the major axis of the conic section relative to the polar reference axis ($\theta = 0$).

Solution: From (1.1.28.1), the general equation of a conic in matrix form is given by

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (1.1.29.2)$$

For a conic with a focus at the origin ($\mathbf{F} = \mathbf{0}$), unit normal vector $\mathbf{n} = \begin{pmatrix} \cos \theta_0 \\ \sin \theta_0 \end{pmatrix}$, eccentricity e , and directrix distance parameter $c = -p$, substituting into (1.1.28.2)–(1.1.28.4) yields:

$$\mathbf{V} = \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top, \quad (1.1.29.3)$$

$$\mathbf{u} = -pe^2 \mathbf{n}, \quad (1.1.29.4)$$

$$f = -p^2 e^2 \quad (1.1.29.5)$$

Expressing $\mathbf{x}$ directly in polar coordinates yields

$$\mathbf{x} = r \mathbf{r}(\theta), \quad \text{where } \mathbf{r}(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} \quad (1.1.29.6)$$

Substituting into (1.1.29.2) yields

$$r^2 (\mathbf{r}(\theta)^\top \mathbf{V} \mathbf{r}(\theta)) + 2r (\mathbf{u}^\top \mathbf{r}(\theta)) + f = 0 \quad (1.1.29.7)$$

Evaluating the inner products:

$$\mathbf{n}^\top \mathbf{r}(\theta) = (\cos \theta_0 \quad \sin \theta_0) \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} = \cos(\theta - \theta_0) \quad (1.1.29.8)$$

$$\mathbf{r}(\theta)^\top \mathbf{V} \mathbf{r}(\theta) = \mathbf{r}(\theta)^\top (\mathbf{I} - e^2 \mathbf{n} \mathbf{n}^\top) \mathbf{r}(\theta) = 1 - e^2 \cos^2(\theta - \theta_0) \quad (1.1.29.9)$$

$$\mathbf{u}^\top \mathbf{r}(\theta) = -pe^2 \cos(\theta - \theta_0) \quad (1.1.29.10)$$

Substituting these into (1.1.29.7):

$$r^2 [1 - e^2 \cos^2(\theta - \theta_0)] - 2rpe^2 \cos(\theta - \theta_0) - p^2 e^2 = 0 \quad (1.1.29.11)$$

Factoring the quadratic expression:

$$[r(1 + e \cos(\theta - \theta_0)) - ep] [r(1 - e \cos(\theta - \theta_0)) + ep] = 0 \quad (1.1.29.12)$$

Taking the positive real root for radial distance $r > 0$:

$$r(1 + e \cos(\theta - \theta_0)) = ep \quad (1.1.29.13)$$

yielding (1.1.29.1)

1.1.30 Gravitational parameter of a body μ is defined as

$$\mu = GM \quad (1.1.30.1)$$

where M is the mass of the body

1.1.31 Specific angular momentum for a satellite with position vector $\mathbf{r}$ and velocity vector $\mathbf{v}$ is

$$h = \|\mathbf{r} \times \mathbf{v}\| \quad (1.1.31.1)$$

1.1.32 Specific Energy for a satellite with position vector $\mathbf{r}$, velocity vector $\mathbf{v}$ and gravitational parameter μ is

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} \quad (1.1.32.1)$$

Where

$$r = \|\mathbf{r}\|, v = \|\mathbf{v}\| \quad (1.1.32.2)$$

1.1.33 The trajectory of a satellite under a central gravitational force is always a conic section.

Solution: The gravitational acceleration $\mathbf{a}$ acting on a satellite is given by Newton's law of gravitation:

$$\mathbf{a} = -\frac{\mu}{r^3} \mathbf{r} \quad (1.1.33.1)$$

Taking the cross product of $\mathbf{r}$ with $\mathbf{a}$:

$$\mathbf{r} \times \mathbf{a} = \mathbf{r} \times \left(-\frac{\mu}{r^3} \mathbf{r} \right) = \mathbf{0} \quad (1.1.33.2)$$

From (1.1.33.2)

$$\frac{d}{dt}(\mathbf{r} \times \mathbf{v}) = \mathbf{v} \times \mathbf{v} + \mathbf{r} \times \mathbf{a} = \mathbf{0} \quad (1.1.33.3)$$

Hence, the specific angular momentum vector $\mathbf{h}$ is conserved:

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} = \text{constant} \quad (1.1.33.4)$$

Crossing the acceleration $\mathbf{a}$ with $\mathbf{h}$:

$$\mathbf{a} \times \mathbf{h} = \left(-\frac{\mu}{r^3} \mathbf{r} \right) \times (\mathbf{r} \times \mathbf{v}) \quad (1.1.33.5)$$

$$= -\frac{\mu}{r^3} [(\mathbf{r} \cdot \mathbf{v}) \mathbf{r} - (\mathbf{r} \cdot \mathbf{r}) \mathbf{v}] \quad (1.1.33.6)$$

$$= \frac{\mu}{r^3} [r^2 \mathbf{v} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{r}] \quad (1.1.33.7)$$

Using $\mathbf{r} \cdot \mathbf{v} = r \frac{dr}{dt}$, this simplifies to:

$$\mathbf{a} \times \mathbf{h} = \mu \left[\frac{\mathbf{v}}{r} - \frac{1}{r^2} \frac{dr}{dt} \mathbf{r} \right] = \mu \frac{d}{dt} \left(\frac{\mathbf{r}}{r} \right) \quad (1.1.33.8)$$

Integrating with respect to time t :

$$\mathbf{v} \times \mathbf{h} = \mu \left(\frac{\mathbf{r}}{r} + \mathbf{C} \right) \quad (1.1.33.9)$$

where $\mathbf{C}$ is the integration constant. Taking the scalar product of (1.1.33.9) with $\mathbf{r}$:

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h}) = \mu \mathbf{r} \cdot \left(\frac{\mathbf{r}}{r} + \mathbf{C} \right) \quad (1.1.33.10)$$

$$\implies (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{h} = \mu (r + rC \cos \theta) \quad (1.1.33.11)$$

$$\implies \|\mathbf{h}\|^2 = \mu r (1 + C \cos \theta) \quad (1.1.33.12)$$

where θ is the angle between $\mathbf{r}$ and $\mathbf{C}$. Rearranging for r yields

$$r = \frac{\frac{\|\mathbf{h}\|^2}{\mu}}{1 + C \cos \theta} = \frac{ep}{1 + e \cos \theta} \quad (1.1.33.13)$$

This is the same equation as (1.1.29.1) where

$$ep = \frac{\|\mathbf{h}\|^2}{\mu} \quad (1.1.33.14)$$

$$\mathbf{e} = \mathbf{C} \quad (1.1.33.15)$$

1.1.34 The squared eccentricity e^2 of a satellite's trajectory is related to its specific orbital energy ε and specific angular momentum magnitude h by

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \quad (1.1.34.1)$$

Solution: From (1.1.33.9) and (1.1.33.15),

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \quad (1.1.34.2)$$

Taking the magnitude squared of both sides using $\|\mathbf{e}\|^2 = \mathbf{e} \cdot \mathbf{e}$:

$$e^2 = \|\mathbf{e}\|^2 = \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \right) \cdot \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \right) \quad (1.1.34.3)$$

Expanding the inner product:

$$e^2 = \frac{\|\mathbf{v} \times \mathbf{h}\|^2}{\mu^2} - \frac{2}{\mu r} [\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h})] + \frac{\mathbf{r} \cdot \mathbf{r}}{r^2} \quad (1.1.34.4)$$

We know that

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h}) = (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{h} = \mathbf{h} \cdot \mathbf{h} = h^2 \quad (1.1.34.5)$$

And,

$$\mathbf{r} \cdot \mathbf{r} = r^2 \quad (1.1.34.6)$$

Substituting these back into (1.1.34.4) yields

$$e^2 = \frac{v^2 h^2}{\mu^2} - \frac{2h^2}{\mu r} + 1 \quad (1.1.34.7)$$

$$= 1 + \frac{2h^2}{\mu^2} \left(\frac{v^2}{2} - \frac{\mu}{r} \right) \quad (1.1.34.8)$$

Substituting (1.1.32.1) into (1.1.34.8) yields:

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \quad (1.1.34.9)$$

1.1.35 Consider the system A system comprising a bar, spring and mass is shown in the figure Fig. 1.1.35.1 The bar, having negligible mass, is made of a material having Young's modulus E , cross-sectional area A and length L . The spring stiffness is k and the mass is M kg. Find the axial stiffness of the bar.

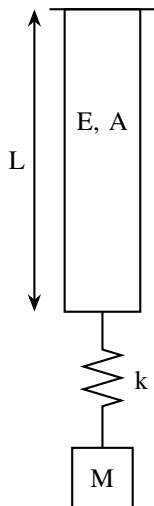


Fig. 1.1.35.1: Spring mass system

Solution: The axial stiffness of the bar is

$$k_b = \frac{AE}{L} \quad (1.1.35.1)$$

1.1.36 Find the overall spring constant in Fig. 1.1.35.1.

Solution: Using Hooke's law, by balancing the displacements,

$$\frac{Mg}{k_t} = \frac{Mg}{k_b} + \frac{Mg}{k} \quad (1.1.36.1)$$

$$\Rightarrow k_{eq} = \frac{k_b k}{k_b + k} \quad (1.1.36.2)$$

where k_t is the overall spring constant.

1.1.37 Show that the equation for motion for the mass in Fig. 1.1.37.1 is

$$\frac{d^2 x}{dt^2} = -\frac{k}{M}x \quad (1.1.37.1)$$

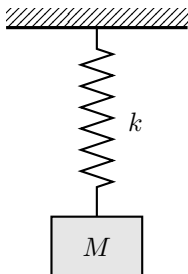


Fig. 1.1.37.1: Spring Mass system in vertical plane

Solution: When the mass M is at rest, let the system stretch by δ_{st} . Then,

$$Mg - k\delta_{st} = 0 \implies Mg = k\delta_{st} \quad (1.1.37.2)$$

When the mass is displaced downward by $x(t)$, using Hooke's law,

$$M \frac{d^2x}{dt^2} = Mg - k(\delta_{st} + x) \quad (1.1.37.3)$$

$$(1.1.37.4)$$

yielding (1.1.37.1), upon substituting from (1.1.37.2).

B. Problems

1.2.1 How many 3-digit numbers can be formed using three distinct single digit prime numbers? (AE 2026)

- i) 64 ii) 24 iii) 12 iv) 4

Solution: There are 4 single digit prime numbers: 2,3,5 and 7. The desired numbers are given by

$${}^4P_3 = \frac{4!}{1!} = 24 \quad (1.2.37.5)$$

The following code verifies the solution

codes/PNC.c

1.2.2 In a group of students, 10 students like Mathematics, 12 students like English, 4 students like both Mathematics and English, and 6 students like neither Mathematics nor English. The number of students in the group is _____. (AE 2026)

- i) 18 ii) 20 iii) 24 iv) 32

Solution: Let N be the total number of students in the group and

- M : The event that the student likes Mathematics.
- E : The event that the student likes English.

The respective probabilities are expressed as

$$\Pr(M) = \frac{10}{N}, \quad \Pr(E) = \frac{12}{N}, \quad \Pr(ME) = \frac{4}{N}, \quad \Pr((M+E)') = \frac{6}{N} \quad (1.2.37.6)$$

From (1.1.2.2)

$$\Pr(M+E) + \Pr((M+E)') = 1 \quad (1.2.37.7)$$

From (1.1.1.1)

$$\Pr(M) + \Pr(E) - \Pr(ME) + \Pr((M+E)') = 1 \quad (1.2.37.8)$$

Substituting from (1.2.37.6) in the above,

$$\frac{10}{N} + \frac{12}{N} - \frac{4}{N} + \frac{6}{N} = 1 \quad (1.2.37.9)$$

$$\Rightarrow N = 24 \quad (1.2.37.10)$$

- 1.2.3 Given a square of side $s = 4\text{cm}$ and two circles, one with known radius $r_1 = 1\text{cm}$ and another with unknown radius r_2 , such that both of them touch each other externally and also the square, find the value of r_2 . Also describe how the figure in the problem can be constructed. (AE 2026)

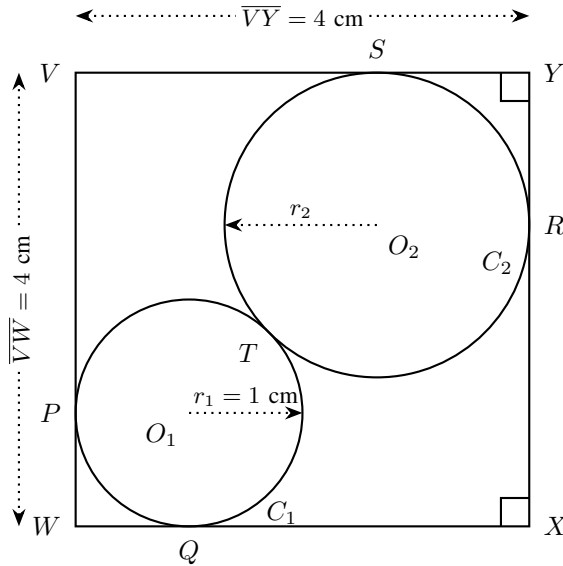


Fig. 1.2.37.2

Solution: In Fig. 1.2.37.2, using (1.1.20.1) and (1.1.20.2),

$$\mathbf{W} = \mathbf{0}, \mathbf{X} = s\mathbf{e}_1, \mathbf{Y} = s\mathbf{1}, \mathbf{V} = s\mathbf{e}_2, \mathbf{O}_1 = r_1\mathbf{1}, \mathbf{O}_2 = (s - r_2)\mathbf{1} \quad (1.2.37.11)$$

Upon substituting from (1.2.37.11)

$$\|\mathbf{O}_2 - \mathbf{O}_1\| = (s - r_1 - r_2)\mathbf{1} = r_1 + r_2 \quad (1.2.37.12)$$

$$\Rightarrow \sqrt{2}(s - r_1 - r_2) = r_1 + r_2 \quad (1.2.37.13)$$

$$\Rightarrow r_2 = \frac{\sqrt{2}s - (\sqrt{2} + 1)r_1}{\sqrt{2} + 1}. \quad (1.2.37.14)$$

Substituting the values of s and r_1 ,

$$r_2 = (7 - 4\sqrt{2})\text{ cm} \quad (1.2.37.15)$$

The following code generates Fig. 1.2.37.3.

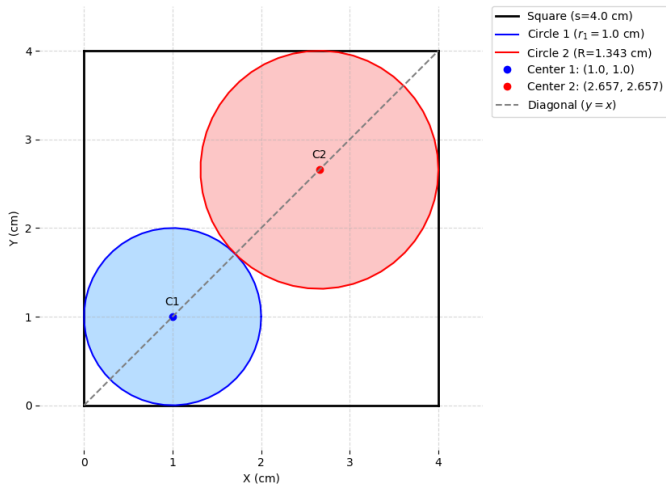


Fig. 1.2.37.3: Square with Tangent Circles

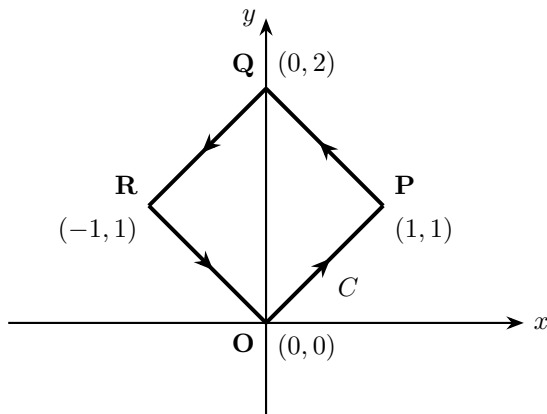
1.2.4 Consider the contour C shown in the figure below. For the vector $\mathbf{F} = (x + 2y)\hat{e}_x + (2x + 4y)\hat{e}_y$, the integral $\oint_C \mathbf{F} \cdot d\mathbf{l} = \underline{\hspace{2cm}}$. Here $d\mathbf{l}$ represents an infinitesimal length along the contour C . (AE 2026)

i) 0

ii) 2

iii) 4

iv) 6

Fig. 1.2.37.4: Contour C

Solution:

$$\therefore \mathbf{F}^\top d\mathbf{l} = (x + 2y)dx + (2x + 4y)dy, \quad (1.2.37.16)$$

$$\int \mathbf{F} \cdot d\mathbf{l} = \int x dx + 2 \int (x dy + y dx) + \int 4y dy \quad (1.2.37.17)$$

$$= \frac{x^2}{2} + 2 \int d(xy) + 2y^2 \quad (1.2.37.18)$$

$$= \frac{x^2}{2} + 2(xy) + 2y^2 \quad (1.2.37.19)$$

$$\therefore \int_{(0,0)}^{(1,1)} \mathbf{F} \cdot d\mathbf{l} = \left[\frac{x^2}{2} \right]_0^1 + 2[xy]_{(0,0)}^{(1,1)} + [2y^2]_0^1 = \frac{9}{2} \quad (1.2.37.20)$$

$$\int_{(1,1)}^{(0,2)} \mathbf{F} \cdot d\mathbf{l} = \left[\frac{x^2}{2} \right]_1^0 + 2[xy]_{(1,1)}^{(0,2)} + [2y^2]_1^2 = \frac{7}{2} \quad (1.2.37.21)$$

$$\int_{(0,2)}^{(-1,1)} \mathbf{F} \cdot d\mathbf{l} = \left[\frac{x^2}{2} \right]_0^{-1} + 2[xy]_{(0,2)}^{(-1,1)} + [2y^2]_2^1 = -\frac{15}{2} \quad (1.2.37.22)$$

$$\int_{(-1,1)}^{(0,0)} \mathbf{F} \cdot d\mathbf{l} = \left[\frac{x^2}{2} \right]_{-1}^0 + 2[xy]_{(-1,1)}^{(0,0)} + [2y^2]_1^0 = -\frac{1}{2} \quad (1.2.37.23)$$

$$\therefore, \oint_C \mathbf{F} \cdot d\mathbf{l} = 0 \quad (1.2.37.24)$$

upon summing up the above integrals. The following code simulates the result. in Fig. 1.2.37.5.

codes/Line_Integral.py

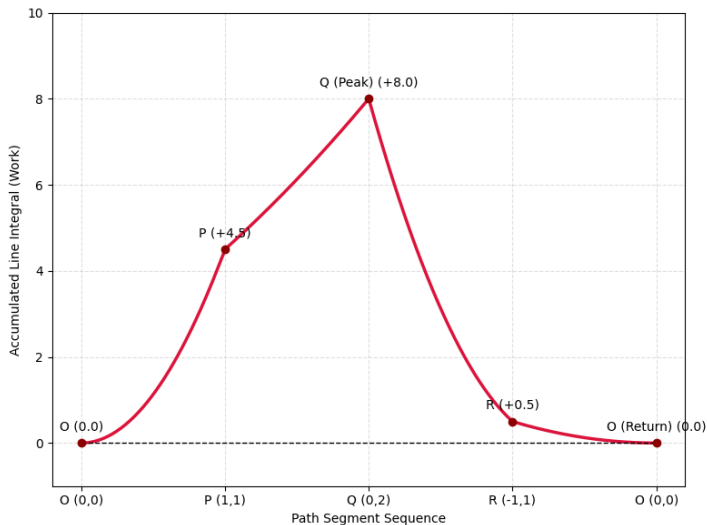


Fig. 1.2.37.5

1.2.5 The Fourier series representation of a square wave is shown in the figure below. The fluctuations seen near $x = \pm 1$ are named after which one of the following scientists? (AE 2026)

- i) Cauchy ii) Fourier iii) Gibbs iv) Laplace

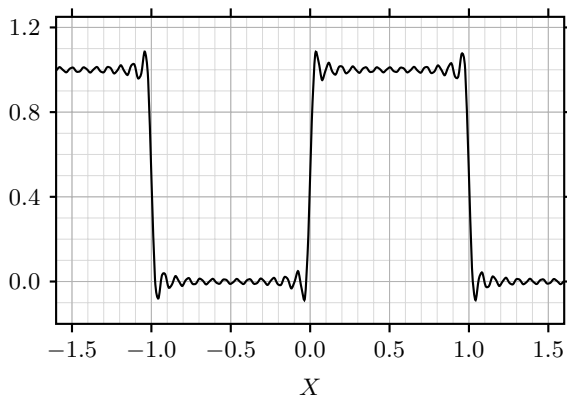


Fig. 1.2.37.6: Gibbs phenomenon

Solution: Fig. 1.2.37.6 can be expressed as a periodic function with parameters

$$x_0(t) = \text{rect} \left(t - \frac{1}{2} \right), T_0 = 2 \quad (1.2.37.25)$$

From (1.1.19.2) and (1.1.18.2), substituting $t_0 = \frac{1}{2}$,

$$c_n = \frac{\sin\left(\frac{\pi k}{2}\right)}{\pi k} e^{-j\pi n} \quad (1.2.37.26)$$

The Gibbs phenomenon is explained in Fig. 1.1.18.1.

- 1.2.6 If a matrix can be written as $\mathbf{A} = \mathbf{u}\mathbf{v}^\top$, where both u and v are n -dimensional real-valued non-zero column vectors, then the rank of the matrix $\mathbf{A}$ is? (AE 2026)

Solution: From (1.1.21.1), the rank is 1.

- 1.2.7 An $n \times n$ square matrix $\mathbf{A}$ satisfies $\mathbf{A}^\top = \mathbf{A}^{-1}$. The determinant of this matrix may take which of the following value(s)? (AE 2026)

- i) 1 ii) -1 iii) n iv) 0

Solution: From the given information,

$$|\mathbf{A}^\top| = |\mathbf{A}^{-1}| \quad (1.2.37.27)$$

$$\Rightarrow |\mathbf{A}| = \frac{1}{|\mathbf{A}|} \quad (1.2.37.28)$$

$$\therefore |\mathbf{A}| = \pm 1 \quad (1.2.37.29)$$

substituting (1.1.22.1) and (1.1.24.1) in (1.2.37.27)

- 1.2.8 For the matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, if the relation $a + b = c + d$ holds and $a, b, c, d \neq 0$, then which one of the following statements about $\mathbf{A}$ is FALSE? (AE 2026)

- i) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector iii) $\lambda = d - b$ is an eigenvalue
ii) $\lambda = a + b$ is an eigenvalue iv) $\lambda = d + b$ is an eigenvalue

Solution: Let

$$a + b = c + d = k \quad (1.2.37.30)$$

Then,

$$\begin{aligned} \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= k \\ \begin{pmatrix} c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= k \end{aligned} \quad (1.2.37.31)$$

yielding

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = k \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.2.37.32)$$

$\therefore \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector and $a + b$ is an eigenvalue from (1.1.25.1). From (1.2.37.30),

$$a - c = d - b = -(b - d) = l \quad (1.2.37.33)$$

Then,

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = l \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (1.2.37.34)$$

Thus, $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigenvector and $d - b$ is the 2nd eigenvalue.

1.2.9 Consider the differential equation:

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = 0 \quad (1.2.37.35)$$

with initial conditions $y|_{x=0} = 0$ and $\left. \frac{dy}{dx} \right|_{x=0} = 1$. Find the value of the slope $\frac{dy}{dx}$ at $x = \ln(2)$ rounded off to three decimal places. (AE 2026)

Solution:

i) Taking the Laplace transform of (1.2.37.35), from (1.1.6.1) and (1.1.7.1),

$$[s^2 Y(s) - sy(0) - y'(0)] + 2[sY(s) - y(0)] + Y(s) = 0 \quad (1.2.37.36)$$

Substituting $y(0) = 0, y'(0) = 1$ in the above,

$$(s^2 Y(s) - 1) + 2sY(s) + Y(s) = 0 \quad (1.2.37.37)$$

$$\implies Y(s) = \frac{1}{(s+1)^2} \quad (1.2.37.38)$$

Taking the inverse Laplace transform, from (1.1.8.1) and (1.1.11.1),

$$y(x) = xe^{-x}, \quad x \geq 0 \quad (1.2.37.39)$$

$$\implies y'(x) = (1-x)e^{-x} \quad (1.2.37.40)$$

$$\text{or, } y'(\ln(2)) = (1 - \ln(2))e^{-\ln(2)} \approx 0.153 \quad (1.2.37.41)$$

ii) From (1.1.13.3) and (1.1.13.6), (1.2.37.35) can be expressed as

$$\frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} + 2 \left(\frac{y_{n+1} - y_n}{h} \right) + y_n = 0 \quad (1.2.37.42)$$

$$\implies y_{n+2} = 2(1-h)y_{n+1} - (1-h)^2 y_n \quad (1.2.37.43)$$

For $h = 0.01$,

$$y_1 = y_0 + h \{y'(0)\} = 0.01 \quad (1.2.37.44)$$

The following code computes (1.2.37.43) and generates Fig. 1.2.37.7.

codes/Euler.py

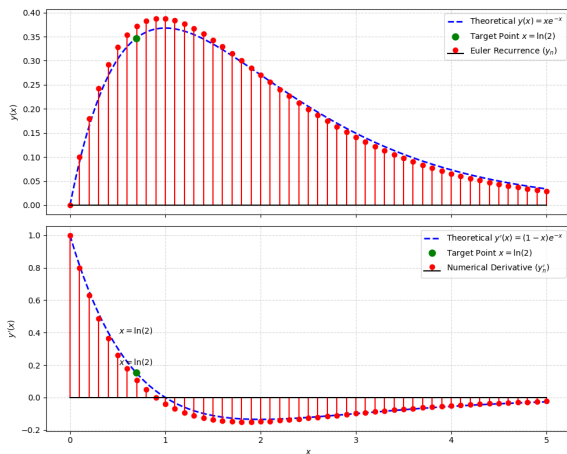


Fig. 1.2.37.7: (1.2.37.39) vs (1.2.37.43)

1.2.38 Find the minimum value of the function $f(x) = |x| + |2x + 3|$ for real x . (AE 2026)

Solution: The following code plots the given function in Fig. 1.2.38.1 and also finds the minimum values.

codes/Modulus_function.py

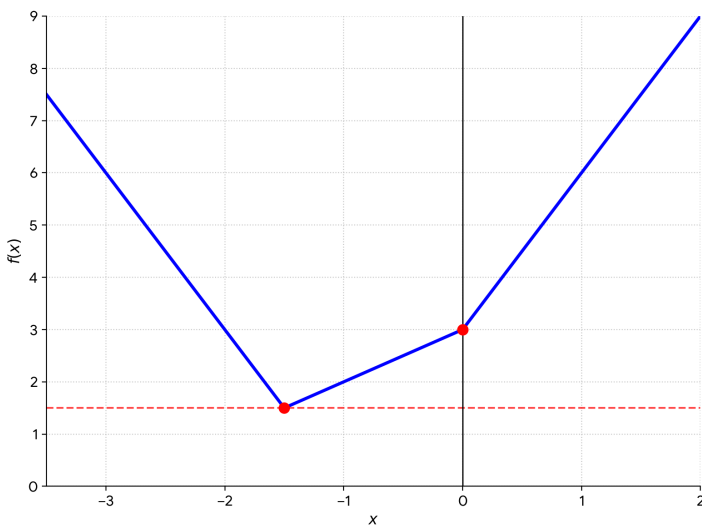


Fig. 1.2.38.1: Plot of the function

1.2.39 Consider the system in Fig. 1.1.35.1. For the following parameters,

- Bar: $E = 200 \text{ GPa}$, $A = 100 \text{ mm}^2$, $L = 100 \text{ mm}$
- Spring Stiffness: $k = 200 \text{ kN/mm}$
- Mass: $M = 100 \text{ kg}$

find the natural frequency of free vibration of the system.

(AE 2026)

Solution: From (1.1.35.1),

$$k_b = \frac{10^{-4} \times 200 \times 10^9}{0.1} = 2 \times 10^8 \text{ N/m} \quad (1.2.39.1)$$

$$k = 200 \text{ kN/mm} = 2 \times 10^8 \text{ N/m} \quad (1.2.39.2)$$

Substituting the above in (1.1.36.2),

$$k_t = \frac{(2 \times 10^8)(2 \times 10^8)}{2 \times 10^8 + 2 \times 10^8} = 10^8 \text{ N/m} \quad (1.2.39.3)$$

From (1.1.37.1) and (1.1.12.6),

$$x(t) = \beta \cos\left(\sqrt{\frac{k_t}{m}}t\right) + \frac{\gamma}{\sqrt{\frac{k_t}{m}}} \sin\left(\sqrt{\frac{k_t}{m}}t\right) \quad (1.2.39.4)$$

$\therefore$ the natural frequency of free vibration is

$$\sqrt{\frac{k_{eq}}{M}} = 1000 \text{ rad/s} \quad (1.2.39.5)$$

upon substituting numerical values. The following code generates Fig. 1.2.39.1.

codes/Spring.py

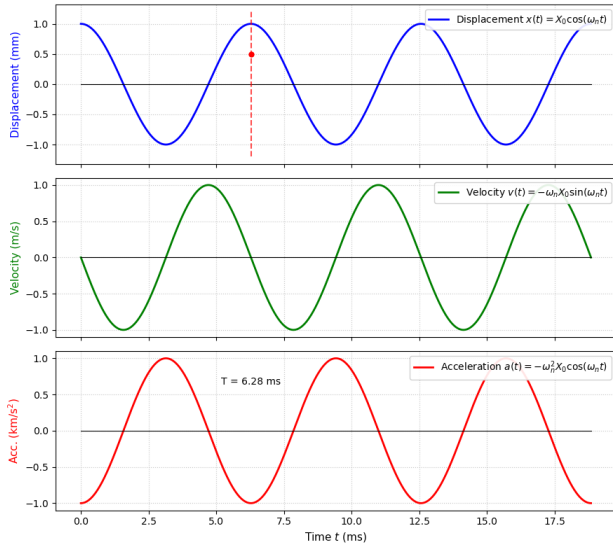


Fig. 1.2.39.1: Acceleration, Velocity and Position of the System

1.2.40 An earth satellite has the instantaneous position vector $\mathbf{r}$ and velocity vector $\mathbf{v}$ as given below. Here $\mathbf{p}$ and $\mathbf{q}$ denote the unit vectors along the x and y axes of the perifocal frame, respectively. Assume that the value of gravitational parameter is $398600 \text{ km}^3/\text{s}^2$. Which one of the following trajectories does the satellite follow? (AE 2026)

$$\mathbf{r} = (8000\mathbf{p} + 9000\mathbf{q}), \text{ km } \mathbf{v} = (-6\mathbf{p} + 6\mathbf{q}) \text{ km/s} \quad (1.2.40.1)$$

- a) Circle b) Hyperbola c) Parabola d) Straight Line

Solution: We are given :

$$\mathbf{r} = \begin{pmatrix} 8000 \\ 9000 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} -6 \\ 6 \end{pmatrix} \quad (1.2.40.2)$$

From (1.1.31.1), Specific Angular Momentum h

$$= |8000(6) - 9000(-6)| = 102000 \text{ km}^2/\text{s} \quad (1.2.40.3)$$

From (1.1.32.1), Specific Energy ε

$$= \frac{72}{2} - \frac{398000}{\sqrt{8000^2 + 9000^2}} \approx 2.898 \quad (1.2.40.4)$$

From (1.1.34.1), eccentricity squared e^2

$$= 1 + \frac{2(2.898)(102000)^2}{(398600)^2} \approx 1 + 0.3795 = 1.3795 \quad (1.2.40.5)$$

Since $e > 1$, the trajectory is a Hyperbola. The following code verifies the solution through a plot

codes/Satellite.py

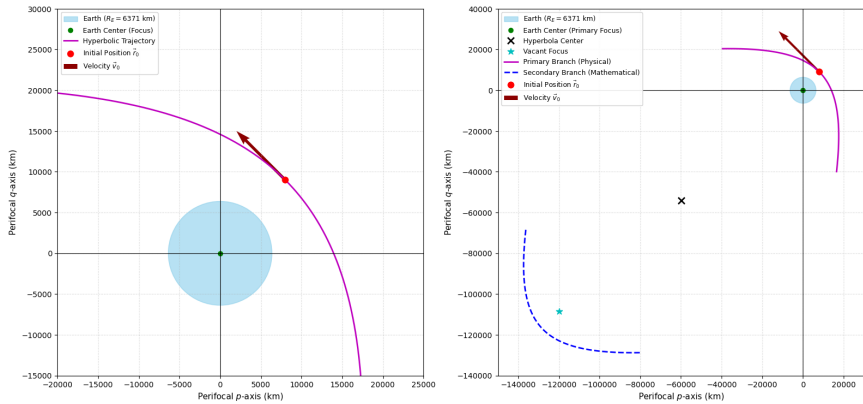


Fig. 1.2.40.1: Trajectory of satellite